THERMO SENSE
Power Budgeting on a Chip

Capstone Project Report
PDEs, Transforms and Optimization Techniques

Submitted by

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Abstract

Thermal management is essential in integrated circuits because uneven power dissipation can create localized hot spots. This project investigates the redistribution of a fixed power budget among four heat-source regions to minimize the maximum steady-state temperature of a simplified chip model. Fourier's law and energy conservation are used to formulate the heat equation, which is discretized using the finite-difference method and solved using sparse linear algebra. Linear programming is then used to determine a power allocation that minimizes the peak temperature while maintaining a total power of 5 W.

The reported simulation gives a peak temperature of 352.91 K for uniform allocation and approximately 342.01 K for optimized allocation. This corresponds to a reduction of about 10.9 K, subject to rounding. The results demonstrate that unequal power distribution can improve the peak-temperature objective within the assumed model. The study also identifies the limitations of steady-state, two-dimensional thermal modelling.

Keywords: Thermal management, Poisson equation, finite differences, sparse matrices, linear programming.

1 Introduction

Modern integrated circuits contain several functional blocks that dissipate electrical power as heat. If the generated heat is not distributed and removed effectively, localized hot spots may increase operating temperature and reduce thermal reliability margins. Uniformly dividing the available power among all blocks is simple, but it does not necessarily minimize the hottest point because the thermal response depends on source location and heat-conduction paths.

The objective of this project is to allocate a fixed power budget among four internal heat sources while minimizing the maximum steady-state temperature. The die is modelled as a two-dimensional square with a side length of 10 mm, a 40×40 grid and a thermal conductivity of 150 W/(m K). All edges are maintained at 300 K, and the total power budget is 5 W.

The uniform baseline assigns 1.25 W to each source. The optimized case permits unequal non-negative source powers while preserving the same total budget. The project combines heat-transfer theory, partial differential equations, transform interpretation, numerical linear algebra and constrained optimization in a single workflow.

2 Mathematical Background and Related Work

Fourier's law describes conductive heat flow, while energy conservation leads to the steady-state heat equation. Numerical methods such as finite differences allow this equation to be solved on a computational grid. Linear programming can then determine an allocation that minimizes the maximum temperature when the temperature response is linear in the source powers.

The project integrates these established techniques rather than relying on a thermal model alone. The governing physics predicts the temperature field, numerical discretization makes the model computationally tractable, and optimization selects a power distribution according to the peak-temperature objective.

3 Methodology: Six Mathematical Stages

3.1 Stage 1: Field Model

Fourier's law expresses the heat-flux vector as

$$\mathbf{q}_h = -k\nabla T, \tag{1}$$

where k is thermal conductivity and T is temperature. At steady state, energy conservation gives

$$\nabla \cdot \mathbf{q}_h = Q, \tag{2}$$

where Q denotes volumetric heat generation. For constant conductivity, these equations combine to give

$$-k\nabla^2 T = Q. \tag{3}$$

3.2 Stage 2: Partial Differential Equation

For a two-dimensional die, the governing equation becomes

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = -\frac{Q}{k}. \tag{4}$$

This is the steady-state Poisson equation. The boundary condition is

$$T(x, y) = 300\,\mathrm{K} \qquad \text{on the die boundary.} \tag{5}$$

The equation describes how internal heat generation produces temperature rise above the fixed boundary temperature.

3.3 Stage 3: Thermal-Impedance Interpretation

Thermal impedance relates temperature rise to applied power:

$$Z_{\mathrm{th}}(s) = \frac{\Delta T(s)}{P(s)}. \tag{6}$$

The limit $s \to 0$ corresponds to the steady-state interpretation when a finite steady response exists. The present simulation solves the steady-state equation directly and does not calculate transient temperature changes.

3.4 Stage 4: Finite-Difference Method

The continuous domain is discretized using a 40×40 grid. The grid spacing is

$$\Delta x = \Delta y = \frac{10\,\mathrm{mm}}{40 - 1} \approx 0.2564\,\mathrm{mm}. \tag{7}$$

For an interior point, the five-point approximation to the Laplacian is

$$\nabla^2 T_{i,j} \approx \frac{T_{i+1,j} + T_{i-1,j} + T_{i,j+1} + T_{i,j-1} - 4T_{i,j}}{\Delta x^2}. \tag{8}$$

The discretized equations are assembled into a sparse linear system,

$$A\boldsymbol{\theta} = \mathbf{b}, \tag{9}$$

where A is the coefficient matrix and $\boldsymbol{\theta}$ contains the unknown temperatures or temperature rises. The system is solved numerically using SciPy sparse linear algebra.

3.5 Stage 5: Linear Programming

Under the assumed constant material properties and boundary conditions, the temperature response is linear in source powers:

$$\mathbf{T} = \mathbf{T}_{\text{amb}} + H\mathbf{p}, \tag{10}$$

where H is the thermal response matrix and $\mathbf{p} = [p_1, p_2, p_3, p_4]^T$.

An auxiliary variable t represents the maximum grid temperature. The optimization problem is

$$\min_{\mathbf{p},t} \quad t \tag{11}$$

$$\text{subject to} \quad T_i(\mathbf{p}) \le t, \quad \forall i, \tag{12}$$

$$\sum_{j=1}^{4} p_j = 5\,\text{W}, \tag{13}$$

$$p_j \ge 0, \quad j = 1, \ldots, 4. \tag{14}$$

The equality constraint ensures that the full power budget is used. SciPy's linear programming solver is used to calculate the allocation.

3.6 Stage 6: Implementation and Verification

The model is implemented in Python using NumPy, SciPy and Matplotlib. The workflow defines the geometry and sources, assembles and solves the thermal system, obtains the optimized power distribution and compares it with the uniform baseline. The total allocated power and non-negativity of the solution are checked to ensure that the optimization satisfies the specified constraints.

4 Experimental Setup

The simulation uses a 10 mm × 10 mm die, a 40 × 40 grid, thermal conductivity of 150 W/(m K), boundary temperature of 300 K and a total power budget of 5 W. Four internal 4 × 4-cell source regions are assumed.

The notebook assumes a die thickness of 0.1 mm. The project brief does not fully specify thickness and source footprints, so these are modelling assumptions rather than measured physical properties. Absolute temperatures depend on these choices.

The baseline allocates 1.25 W to every source. The optimized experiment redistributes the same total power according to the linear programming solution. Both cases use the same thermal model and boundary conditions, allowing their reported peak temperatures to be compared.

5 Results and Discussion

The reported numerical results are summarized in Table 1.

Table 1: Uniform and optimized power allocation

Quantity	**Uniform**	**Optimized**
Source 1 power	1.250 W	0.801 W
Source 2 power	1.250 W	1.364 W
Source 3 power	1.250 W	1.417 W
Source 4 power	1.250 W	1.417 W
Total power	5 W	Approximately 5 W
Peak temperature	352.91 K	342.01 K

The peak-temperature reduction calculated from the displayed temperatures is

$$\Delta T_{\max} = 352.91 - 342.01 = 10.90\,\mathrm{K}. \tag{15}$$

The simulation summary reports approximately 10.89 K; the small difference is attributable to rounding. The relative reduction based on the displayed temperatures is

$$R_T = \frac{352.91 - 342.01}{352.91} \times 100 \approx 3.09\%. \tag{16}$$

The optimized allocation is unequal. The first source receives less power than in the uniform case, while the other three receive more. This indicates that equal sharing is not necessarily the allocation that minimizes the maximum temperature in the assumed model. The optimizer considers the combined temperature effect of all source powers rather than attempting to equalize power among the blocks.

The main inference is that power distribution can influence the peak temperature even when the total power remains fixed. The reported result also demonstrates how a physical model, numerical solution and optimization algorithm can work together to produce a constrained allocation. The result is model-specific and does not, by itself, establish the same improvement in a physical chip.

6 Limitations and Future Scope

The model is two-dimensional and assumes steady-state conditions. It therefore cannot represent transient power spikes, thermal storage or heat flow through a complete three-dimensional chip package. Thermal conductivity is treated as constant, and all edges are fixed at 300 K, which idealizes the cooling environment. The source footprints and die thickness are also assumed rather than derived from a specific physical layout.

Future work can include a time-dependent heat equation, three-dimensional geometry, temperature-dependent material properties and realistic cooling boundaries. A grid-convergence study would help quantify numerical sensitivity. Further constraints, such as minimum and maximum power per block, could also be introduced to make the allocation more representative of real operating conditions.

7 Conclusion

This project combines Fourier's law, the steady-state Poisson equation, thermal-impedance interpretation, finite-difference discretization, sparse linear algebra and linear programming to study thermal-aware power budgeting. The reported simulation reduces the peak temperature from 352.91 K under uniform allocation to approximately 342.01 K under optimized allocation, while maintaining a total budget of approximately 5 W.

The reported improvement is about 10.9 K, or approximately 3.09% relative to the displayed uniform peak temperature. The result shows that unequal power allocation can improve the peak-temperature objective within the chosen model. However, the conclusion depends on the assumed geometry, thickness, material properties, grid and boundary conditions. Further physical validation and more realistic thermal modelling are required before applying the numerical result directly to hardware.

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